

# Cosmology

## Exercises Sheet 4

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### Exercise 1

The flat  $\Lambda$ CDM model is a mathematical model of the Big Bang theory with a cosmological constant ( $\Lambda$ ) associated with dark energy, the postulated cold dark matter, and the ordinary matter. In this model we have the matter density parameter  $\Omega_{m,0} = 0.3$  and the Hubble parameter  $h = 0.7$ . Let's start to plot the cosmological distances (in Gpc) in function of redshift.

The Hubble law's distance can be obtained from a simple approximation relation between redshift  $z$  and the proper distance  $d_p$ . We know that:

$$z = \frac{R(t_0)}{R(t_E)} - 1 \quad (1)$$

where  $t_E$  is the time of photons emission and  $t_0$  is the time of observation. Taylor expanding  $z$ , we obtain:

$$\begin{aligned} z &= \left( \frac{R(t_0)}{R(t_E)} - 1 \right)_0 + \frac{d}{dt_E} \left( \frac{R(t_0)}{R(t_E)} - 1 \right)_0 (t_E - t_0) + \dots \\ &\approx - \left( \frac{R(t_0)}{R^2(t_E)} \dot{R}(t_E) \right)_0 (t_E - t_0) = \frac{\dot{R}(t_E)}{R(t_0)} (t_0 - t_E) = H_0(t_0 - t_E) \end{aligned} \quad (2)$$

Now we want to replace  $(t_0 - t_E)$  with a distance and to do this we assume that if the distance is not too large the time interval is simply the proper distance  $d_p$  divided by the speed of light:

$$z \approx H_0(t_0 - t_E) \approx H_0 \frac{d_p}{c} \quad (3)$$

Therefore, we obtain a simplified Hubble-law distance as a function of redshift:

$$d_p \approx \frac{cz}{H_0} \quad (4)$$

where  $H_0 = 100h$ . This approximation is only valid for nearby sources. Using the Eq (4), we obtain the plot of Hubble-law distance, as shown in Fig.(1). Hubble's law says simply that

a galaxy's distance is directly proportional to its redshift, or its velocity, so the more distant galaxies are moving faster away from us. Hubble's law also tells us something important about the state of the universe. If the universe is static and unchanging, there should be no correlation between distance and velocity. However, if the universe is expanding, we expect a correlation between distance and velocity and this is actually what we observe.

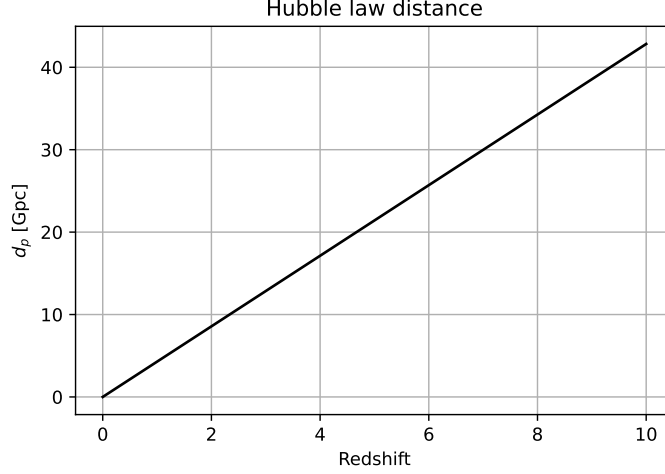


Figure 1: Hubble-law distance as a function of redshift for a flat  $\Lambda$ CDM model with  $\Omega_{m,0}=0.3$  and  $h=0.7$ .

The travel-time distance is simply defined as the integral of  $c$  in time from the time of photons emission to the time of photons observation:

$$d_T = \int_{t_E}^{t_0} c dt = c \int_{R_E}^{R_0} \frac{dR}{\dot{R}} = c \int_{R_E}^{R_0} \frac{dR}{R H_0 E(z)} = \frac{c}{H_0} \int_0^{z_E} \frac{1}{(1+z)E(z)} dz \quad (5)$$

where we have used the first Friedmann Eq.  $H^2 = H_0^2 E^2(z)$  with  $E(z)$  defined as:

$$E^2(z) = \sum_i \Omega_{i,0} (1+z)^{3(1+\omega_i)} \quad (6)$$

and since we are in a  $\Lambda$ CDM model we have only  $\omega_m = 0$  and  $\omega_\Lambda = -1$ , so:

$$E(z) = \sqrt{\Omega_{m,0}(1+z)^3 + (1 - \Omega_{m,0})} \quad (7)$$

This distance  $d_T$  is the time that it took light to reach the observer from the object multiplied by the speed of light. Indeed, the light travel time is also model-dependent result, but it is not an observable. The plot is shown in Fig.(2). The travel-time distance increase with redshift until it reaches a maximum value for high redshift.

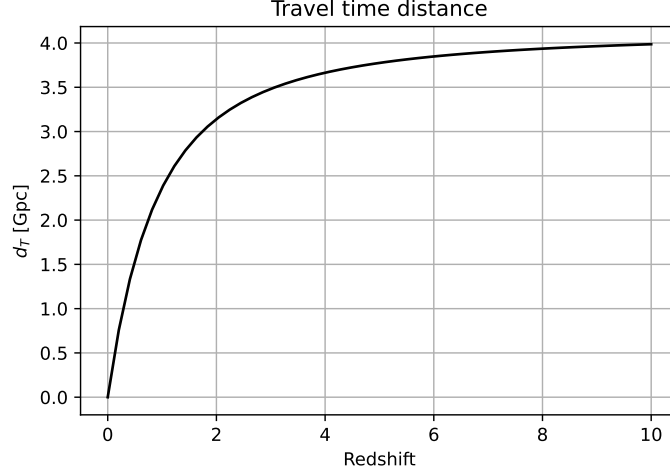


Figure 2: Travel-time distance as a function of redshift for a flat  $\Lambda$ CDM model with  $\Omega_{m,0}=0.3$  and  $h=0.7$ .

Before defining the comoving distance, let's recall the proper distance. This is the distance between two events happening at constant cosmic time. It is the physical distance, and it is impossible to measure as it is defined only at one particular moment in time. Starting from FRW metric, it's possible obtain the proper distance:

$$d_p = R(t) \int_0^{x_E} \frac{dx}{\sqrt{1-kx^2}} = R(t)f(x_E) \quad (8)$$

with

$$f(x_E) = \begin{cases} x_E & k = 0 \\ \frac{1}{\sqrt{|k|}} \arcsin \sqrt{|k|x_E} & k = 1 \\ \frac{1}{\sqrt{|k|}} \operatorname{arcsinh} \sqrt{|k|x_E} & k = -1 \end{cases} \quad (9)$$

The comoving distance is defined as proper distance at some pre-defined reference time. For convenience we use today's time as reference. Hence, the comoving distance is

$$d_c = R(t_0)f(x_E) \quad (10)$$

If setting  $R(t_0) = 1$ , then  $f(x_E)$  is the comoving distance. It's important to note that the comoving distance does not change with time. Connecting  $f(x_E)$  with the redshift, the comoving distance become:

$$d_c = \frac{c}{H_0} \int_0^{z_E} \frac{1}{E(z)} dz \quad (11)$$

Using this last equation, we plot the comoving distance as shown in Fig.(3). In the plot we observe at the beginning a curve that it's an increasing function of the redshift and then it reaches a point where it becomes asymptotic to a maximum value of distance (as you can see better in the Fig.(6)). This trend is explained by Hubble's law, which tells us that when the Universe was younger the expansion was faster, therefore also the comoving distance. In

particular the comoving distance does not increase linearly with the redshift because it depends on the integral of  $(1+z)^{-3}$ . Hence, the comoving distance reaches a maximum beyond which the effect of the expansion is no longer dominant over other cosmological factors, so there is a limit to the observable universe, bounded by the cosmological horizon.

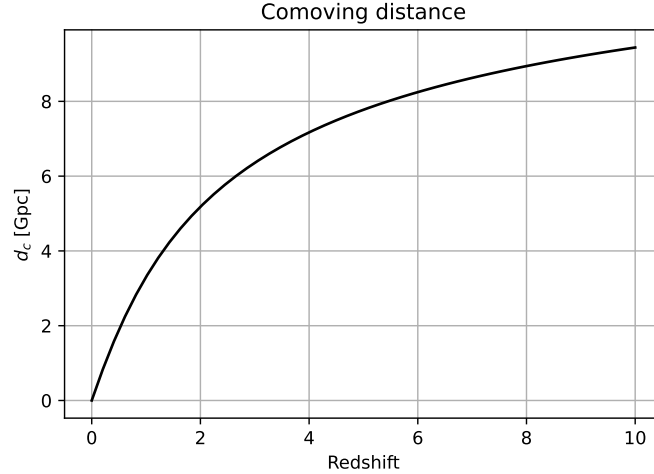


Figure 3: Comoving distance as a function of redshift for a flat  $\Lambda$ CDM model with  $\Omega_{m,0} = 0.3$  and  $h = 0.7$ .

The luminosity distance is defined by the relationship between the observable flux  $F_{obs}$  and the emitted luminosity  $L_E$ :

$$d_L = \sqrt{\frac{4\pi F_{obs}}{L_E}} \quad (12)$$

Then, in term of redshift the luminosity distance is defined as:

$$d_L = (1+z)R_0 x_E \quad (13)$$

where for the  $\Lambda$ CDM model ( $k=0$ )  $x_E$  is:

$$x_E = \frac{d_c}{R_0} \quad (14)$$

This quantity is important for measurements of standard candles like type Ia supernovae, which were first used to discover the acceleration of the expansion of the universe. Replacing the Eq.(14) in Eq.(13), we use the Eq.(13) to obtain the plot shown in Fig.(4). We observe that the luminosity distance is an increasing function of redshift. At large redshift the luminosity distance tends to grow like the Hubble-law distance, so one might think that the proposed analytical approximation of the luminosity distance in the flat  $\Lambda$ CDM model can be reliable up to  $z \sim 2.5$ .

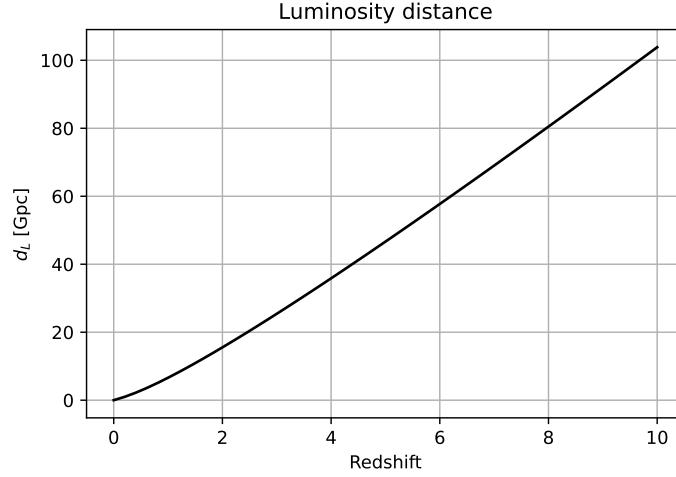


Figure 4: Luminosity distance as a function of redshift for a flat  $\Lambda$ CDM model with  $\Omega_{m,0}=0.3$  and  $h=0.7$ .

The angular diameter distance is defined as:

$$d_A = \frac{D}{\theta_{obs}} = R(t)x_E \quad (15)$$

where for the  $\Lambda$ CDM model ( $k = 0$ )  $x_E$  is:

$$x_E = \frac{d_c}{R_0} \quad (16)$$

Using Eq.(15) we obtain the plot shown in Fig.(5). We observe that we have a maximum, after which the value of the angular distance decreases as the redshift increases. This maximum value for the angular distance of any object implies a minimum solid angle. So with the reasonable assumption that there is a minimum cross-sectional area for any real physical object, we find that the number of objects an observer can see is limited, in principle. Moreover, at redshifts higher than  $z \sim 1.61$ , the angular diameter distance diminishes because the scale factor (“size” of the Universe) is smaller at the moment of time when light from a remote source is emitted than at the moment of time when this light is detected by the observer.

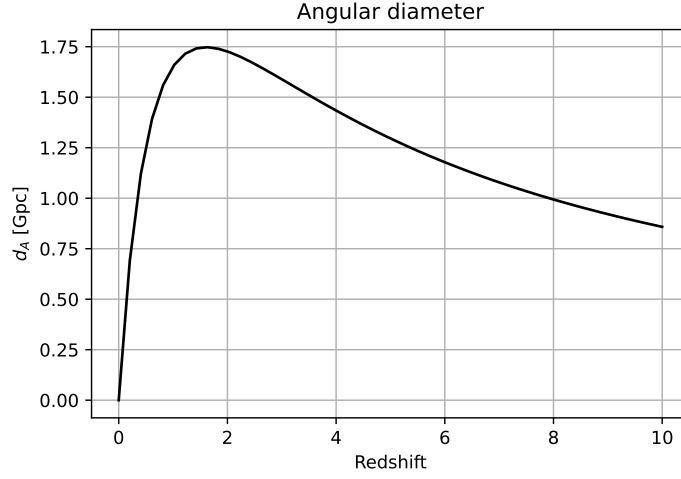


Figure 5: Angular diameter distance as a function of redshift for a flat  $\Lambda$ CDM model with  $\Omega_{m,0}=0.3$  and  $h=0.7$ .

We can see that the different definitions of "distance" in cosmology are all asymptotic one to another for low redshifts (right panel of Fig.(6)). All plot are done numerically in Python (the code is in Appendix A) and the redshift interval on x-axis is chosen considering that in  $\Lambda$ CDM model the scalar field becomes important at late-times, indeed at these late times radiation can be ignored. Finally, we summarize in one plot all cosmological distances discussed, in Fig.(6).

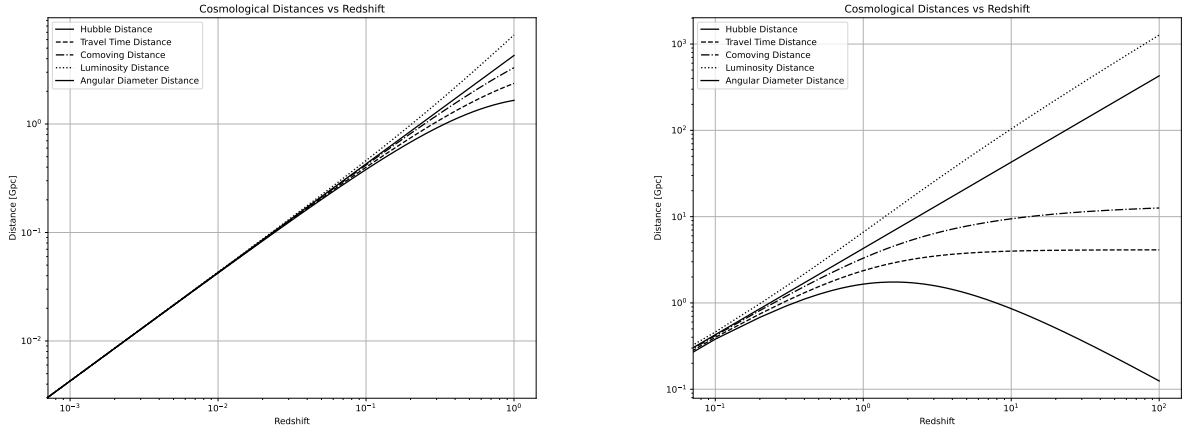


Figure 6: On the left a comparison of cosmological distance measures, from redshift zero to redshift of 1. On the right a comparison of cosmological distance measures, from redshift zero to redshift of 100 to show what the various curves tend to as the redshift increases. The two plots are in scale log-log.

## Exercise 2

The surface brightness of an astronomical object is defined as its observed flux divided by its observed angular scale:

$$\Sigma = \frac{F_{obs}}{\theta_{obs}^2} \quad (17)$$

The observed flux is defined as its brightness per unit area:

$$F_{obs} = \frac{L_E}{4\pi d_L^2} \quad (18)$$

The observed angular scale is defined as object's dimension divided by the angular diameter distance:

$$\theta_{obs} = \frac{D}{d_A} \quad (19)$$

Therefore, we can write the evolution of surface brightness in function of redshift  $z$ :

$$\Sigma = \frac{L_E}{4\pi d_L^2} \frac{d_A^2}{D^2} = \frac{1}{4\pi} \frac{L_E}{D^2} \left( \frac{\frac{1}{1+z} R_0 x_E}{(1+z) R_0 x_E} \right)^2 = \frac{1}{4\pi} \frac{L_E}{D^2} \frac{1}{(1+z)^4} \quad (20)$$

where  $L_E$  and  $D$  can be measured observationally, in particular  $L_E$  requires standard candle and  $D$  requires standard ruler. We can observe that the surface brightness drops as  $1/(1+z)^4$ . In a static Euclidean space, it can easily verify that the surface brightness is independent of distance. The reason is simple: both flux and solid angle (angular area) drop as the inverse of the square of the distance, so their ratio remains constant. Moreover, we have compute the redshift evolution of  $\Sigma(z)$  for a class of objects that are both standard candle and rule, so we can observe that the surface brightness of such objects is independent of any cosmological model.

## Exercise 3

The exercise asks to plot the redshift evolution of the mass of collapsing halos, i.e  $M_{\nu\sigma}(z)$ , varying both the primordial power spectrum index  $n$  and the parameter  $\nu$  defined via the equation:

$$\nu = \frac{\delta_c}{D(z)\sigma(M_{\nu\sigma})} \quad (21)$$

where  $D(z)$  is the the linear growth factor and  $\delta_c \simeq 1.69$  is a critical density used in the Press–Schechter formalism, where there is the assumption that the halos are spherical and virialized overdensities with a density above a critical density  $\delta_c$ . We start to find an analytical expression for mass variance  $\sigma^2(M)$ , that it's defined as:

$$\sigma^2(M) = \frac{1}{2\pi^2} \int_0^\infty P_0(k) \hat{W}^2(kR) k^2 dk \quad (22)$$

where the power spectrum  $P_0(k)$  is taken as a power law:

$$P_0(k) = Ak^n \quad (23)$$

and where the *window function* (or "*filter*")  $\hat{W}(x)$  is defined follow the *Top-Hat filter* as:

$$\hat{W}(x) = \frac{3}{x^3}(\sin(x) - x \cos(x)) \quad (24)$$

Therefore, the Eq.(22) becomes:

$$\begin{aligned} \sigma^2(M) &= \frac{1}{2\pi^2} \int_0^\infty P_0(k) \hat{W}^2(kR) k^2 dk \\ \sigma^2(M) &= \frac{1}{2\pi^2} \int_0^\infty Ak^n \frac{9}{(kR)^6} (\sin(kR) - kR \cos(kR))^2 k^2 dk \\ \sigma^2(M) &= \frac{1}{2\pi^2} \int_0^\infty Ak^{n-4} \frac{9}{R^6} (\sin(kR) - kR \cos(kR))^2 dk \end{aligned} \quad (25)$$

Now, using the following substitution:  $\tilde{k} \equiv Rk$  we obtain:

$$\begin{aligned} \sigma^2(M) &= \frac{A}{2\pi^2} \int_0^\infty \left(\frac{\tilde{k}}{R}\right)^{n-4} \frac{9}{R^6} (\sin(\tilde{k}) - \tilde{k} \cos(\tilde{k}))^2 \frac{d\tilde{k}}{R} \\ \sigma^2(M) &= \frac{A}{2\pi^2} \int_0^\infty \tilde{k}^{n-4} R^{-n+4-6-1} 9 (\sin(\tilde{k}) - \tilde{k} \cos(\tilde{k}))^2 d\tilde{k} \\ \sigma^2(M) &= \frac{A}{2\pi^2} \int_0^\infty \tilde{k}^{n-4} R^{-n-3} 9 (\sin(\tilde{k}) - \tilde{k} \cos(\tilde{k}))^2 d\tilde{k} \end{aligned} \quad (26)$$

Now, relating  $R$  with mass  $M$  by the relation:

$$M = \frac{4\pi}{3} \Omega_m \rho_{crit} R^3 \implies R = \left( \frac{3M}{4\pi \Omega_m \rho_{crit}} \right)^{\frac{1}{3}} \quad (27)$$

we obtain:

$$\sigma^2(M) = \frac{A}{2\pi^2} \left( \frac{3M}{4\pi \Omega_m \rho_{crit}} \right)^{-\frac{n+3}{3}} \int_0^\infty \tilde{k}^{n-4} 9 (\sin(\tilde{k}) - \tilde{k} \cos(\tilde{k}))^2 d\tilde{k} \quad (28)$$

Since we know that  $\Omega_m = \frac{\bar{\rho}}{\rho_{crit}}$ :

$$\begin{aligned} \sigma^2(M) &= \left( \frac{4\pi \bar{\rho}}{3} \right)^{\frac{n+3}{3}} M^{-\frac{n+3}{3}} \frac{A}{2\pi^2} \int_0^\infty \tilde{k}^{n-4} 9 (\sin(\tilde{k}) - \tilde{k} \cos(\tilde{k}))^2 d\tilde{k} \\ \sigma^2(M) &= \left( \frac{4\pi \bar{\rho}}{3} \right)^{\frac{n+3}{3}} M^{-\frac{n+3}{3}} \sigma_0^2 = \left( \frac{M}{M_*} \right)^{-\frac{n+3}{3}} \\ \sigma(M) &= \left( \frac{M}{M_*} \right)^{-\frac{n+3}{6}} \end{aligned} \quad (29)$$

with  $M_* = \sigma_0^{\frac{6}{3+n}} \frac{4\pi\bar{\rho}}{3}$ , which can be assume to be  $M_* = 10^{13} M_\odot$ . For a top-hat window, the above relation hold only for  $-3 < n < 1$ . The integration in Eq.(22) diverges in the limits  $k \rightarrow 0$  and  $k \rightarrow \infty$  for  $n \leq -3$  and  $n \geq 1$ , respectively, because  $\hat{W}^2(kR) \simeq 1$  for  $kR \ll 1$ , and  $\hat{W}^2(kR) \propto k^{-4}$  for  $kR \gg 1$ . The constraint that  $n > -3$  for small  $k$  (large length scale) is required because otherwise the perturbation amplitude would become larger and larger as the wavelength of the perturbation mode increases. The divergence at large  $k$  (short length scale) for  $n \geq 1$  is due to the sharp edge of the top-hat window, which causes the Fourier transform of the window function to die off too slowly as  $k \rightarrow \infty$  to suppress the small-scale power implied by  $n \geq 1$ . Since the mass variance in a top-hat window is a well-defined quantity and should be finite for any realistic density field, the above argument implies that the effective spectral index at  $k \rightarrow \infty$  must be smaller than 1 for any realistic power spectrum. Now, we put the last Eq.(29) of  $\sigma(M)$  in (21):

$$\begin{aligned} \nu &= \frac{\delta_c}{D(z) \left( \frac{M_{\nu\sigma}}{M_*} \right)^{-\frac{n+3}{6}}} \\ \nu &= \frac{\delta_c}{D(z)} \left( \frac{M_{\nu\sigma}}{M_*} \right)^{\frac{n+3}{6}} \\ \frac{M_{\nu\sigma}}{M_*} &= \left( \frac{\nu D(z)}{\delta_c} \right)^{\frac{6}{n+3}} \end{aligned} \quad (30)$$

Now, we consider that the growth factor can be approximated as:

$$D(a = 1/(1+z)) = \frac{5a}{2} \Omega_m(a) \left[ \Omega_m^{4/7}(a) - \Omega_\Lambda(a) + \left( 1 + \frac{\Omega_m(a)}{2} \right) \left( 1 + \frac{\Omega_\Lambda(a)}{70} \right) \right]^{-1} \quad (31)$$

Then, using  $\Omega_{m,0} = 1$  and  $\Omega_{\Lambda,0} = 0$  we can compute the evolution of density parameter as:

$$\Omega_i(z) = \frac{\Omega_{i,0}}{\Omega_{m,0}(1+z)^3 + \Omega_{\Lambda,0}} (1+z)^{3(1+\omega_i)} \quad (32)$$

So, with  $\omega_m = 0$  we find  $\Omega_m = 1$  and with  $\omega_\Lambda = -1$  we find  $\Omega_\Lambda = 0$ . Finally, the mass  $M_{\nu\sigma}$  as function of redshift  $z$  is:

$$M_{\nu\sigma}(z) = M_* \left( \frac{\nu}{\delta_c(1+z)} \right)^{\frac{6}{n+3}} \quad (33)$$

Using the Python program, the Eq.(33) is plotted numerically. The result is shown in Fig. (7). Three curves (representing a group) are depicted, with different values of  $\nu$  for fixed  $n$ , and each group is associated a different value of  $n$ . In particular, we have set 4 values of  $n$  (4 groups) such that  $n$  takes on the following values:  $n = \{-2.5, -2, -1, 0.8\}$  for the reason that we said before. While we have chosen the following values of  $\nu = \{2, 3, 4\}$  because the formation of first proto-galaxies are most likely between redshift  $20 < z < 50$ .

Then we compare the  $M_{\nu\sigma}(z)$  with the Jeans mass  $M_J$ , defined as:

$$M_J = 4.9 \cdot 10^4 \cdot (\Omega_{m,0} h^2)^{-1/2} \left( \frac{1+z}{150} \right)^{3/2} \quad (34)$$

where it's clear that  $M_J$  grows with the redshift. We check where they cross and this tell us at what redshift can a dark matter halo collapse. Therefore, from our numerical simulation, it is observed that to form proto-galaxies in redshift range  $20 < z < 50$ , the most probable values of  $n$  are between  $-2.5 \leq n \leq -2$ . For example with  $n = -2$  and  $\nu = 2$  the formation of proto-galaxies happens at redshift  $z \approx 40$ .

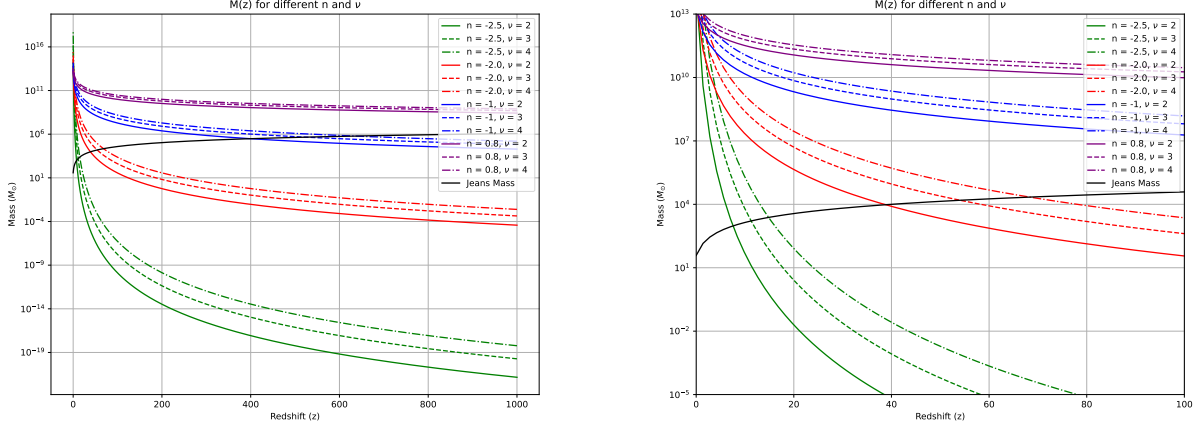


Figure 7: On the right there is the redshift evolution of  $M_{\nu\sigma}$  until  $z = 1000$ . On the left there is an enlargement, so the redshift evolution of  $M_{\nu\sigma}$  until  $z = 100$ .

## Exercise 4

Gravitational waves have two independent polarization states denoted  $+$  and  $\times$ ; a general signal can be described as a linear combination of the two polarization states,  $h = A_+ h_+ + A_\times h_\times$ . The sensitivity of a detector to these depends upon the relative orientations of the source and detector. The output of a gravitational wave detector  $s(t)$  contains a superposition of noise  $n(t)$  and (possibly) a signal  $h(t)$ ,

$$s(t) = n(t) + h(t) \quad (35)$$

For sources with a well-defined  $h(t)$ , one can accumulate the power in the signal over many cycles  $N$ , however for sources of a stochastic nature, we do not have a well-defined  $h(t)$ . For the various detectors, their sensitivities are defined by the power spectral density of their noise  $S_n(f)$ . In our case,  $S_n(f)$  is defined as  $S_n(f) = 0.00938426 \cdot f^{-20}$  sec. We are also given the maximum frequency of observations that is  $f_{max} = 200$  Hz. The signal-to-noise ratio ( $S/N$ ) of the observation is  $S/N \approx 8$ . Finally, we are given a range of frequencies as a function of time (see Fig. 8 left panel).

First of all, we plot the frequencies on a graph and make several fits to see which was the best fit. If we imagine the observed system as an interaction between massive objects (such as black holes) then we expect the frequency to increase until they merge. This is what the data tell us. From here we'll call the time when the two object merge, the time coalescence  $t_c$ . At first glance, one could fit the data with a straight line, i.e. with a function of the type  $m(\mathcal{M}_c)t + q$

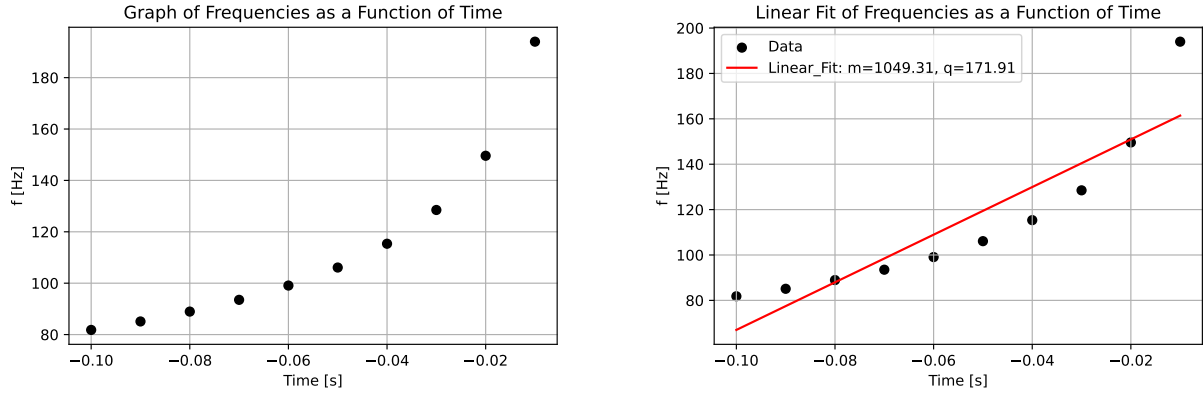


Figure 8: On the left is the frequency trend as a function of time. On the right, a linear fit was performed on the data.

where  $m(\mathcal{M}_c)$  is the slope of the line (which depends on the chirp mass  $\mathcal{M}_c$ ) and  $q$  the intercept (this fit is shown in Fig. (8) right panel). However, it is not a good fit because the chi-square value  $\chi^2$  for the fit of those data to a straight line is  $\chi^2 = 15.92$ , it indicates a discrepancy between the observed data and the values predicted by the linear model.

Clearly this is due to the fact that the data does not follow a straight line, but looks more like a power law. Indeed, one only has to plot the graph on a logarithmic scale to realise that in this case it is a straight line, i.e. translated to a non-logarithmic scale, it is indeed a power law. At this point we can then perform another fit that is no longer linear, but of the type  $m(\mathcal{M}_c) \cdot x^\alpha + q$  and find the values of  $m(\mathcal{M}_c)$ ,  $\alpha$  and  $q$ . This is what was done in Fig. (9) left panel. What we are really interested in in this fit is above all  $\alpha$ . We obtain an  $\alpha$  value of  $-0.375$  which translated into fractional terms is  $-3/8$ . However, from theory, we know that there is a precise law that relates frequency to time. In fact, it can be demonstrated from the following differential equation:

$$\mathcal{M}_c = \frac{c^3}{G} \left( \left( \frac{5}{96} \right)^3 \pi^{-8} f_{GW}^{-11} \dot{f}_{GW}^3 \right)^{\frac{1}{5}} \quad (36)$$

the solution is simply:

$$f_{GW}^{-\frac{8}{3}}(t) = \frac{(8\pi)^{\frac{8}{3}}}{5} \left( \frac{G\mathcal{M}_c}{c^3} \right)^{\frac{5}{3}} (t_c - t) \quad (37)$$

where  $t_c$  is the time coalescence. The time coalescence refers to the moment when two objects in close orbit, such as two black holes or two neutron stars, come so close that they begin to merge together. In the contextualisation of gravitational waves, temporal coalescence refers to the precise moment when two objects are about to come into physical contact and merge, emitting an intense amount of energy in the form of gravitational waves. We then performed the fit with this function to find the chirp mass. The chirp mass is important because it provides a combined measure of the masses of the objects involved, which is directly reflected in the characteristics of the gravitational wave emitted during fusion. For example, the chirp mass influences the frequency of the chirp phase of the gravitational wave, which is a key characteristic

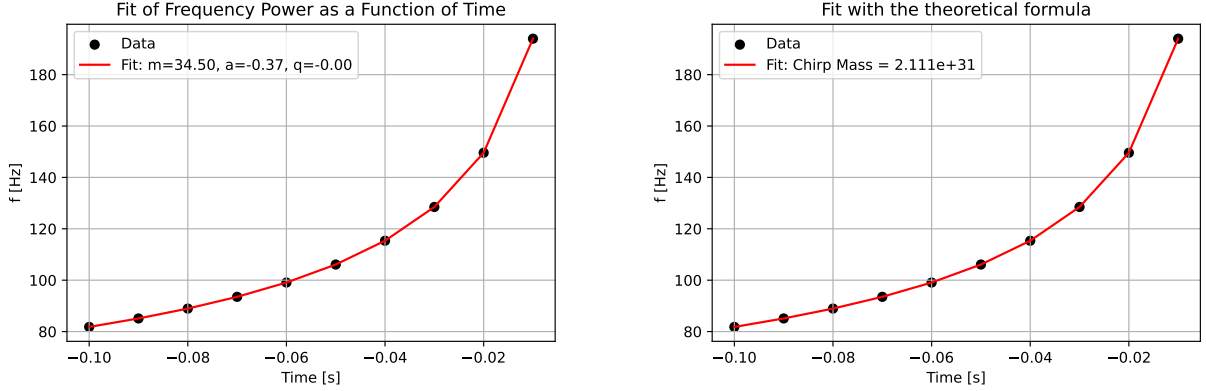


Figure 9: On the left the fit with a generic power law, and on the right the fit using Eq. (37). As can be seen, the two fits are practically identical.

used to identify and characterise gravitational wave sources. The fit is practically identical to that of power law since Eq. (37) is to all intents and purposes a power law. From the fit, we obtain that:

$$\mathcal{M}_c \approx 2.11 \cdot 10^{31} \text{ kg} \quad (38)$$

which corresponds to approximately<sup>1</sup>  $\mathcal{M}_c = 10.55 M_\odot$ .

The extraordinary thing we can derive beyond the chirp mass is the distance of the system. It is possible to average the distances according to the inclination, thus obtaining the following formula:

$$d_{\text{ight}} = \frac{2}{5} \left(\frac{5}{6}\right)^{\frac{1}{2}} \frac{c}{\pi^{2/3}} \left(\frac{G\mathcal{M}_c}{c^3}\right)^{\frac{5}{6}} \left[ \int_0^{f_{\text{max}}} df \frac{f^{-\frac{7}{3}}}{S_n(f)} \right]^{\frac{1}{2}} \left(\frac{S}{N}\right)^{-1} \quad (39)$$

Since we all know the various constituents of this formula, all we have to do is solve it and find the numerical value. In the end we obtain:

$$d_{\text{ight}} = 1.234 \cdot 10^{25} \text{ m} \quad (40)$$

which translated into Mpc is roughly  $d_{\text{ight}} = 400 \text{ Mpc}$ .

Having the distance and chirp mass, we can finally plot the reconstructed strains as a function time for the last 0.1 secs before the merger (assuming the inclination angle equal to  $\pi/3$ ). As mentioned earlier, we are considering an interaction between two compact objects, so for example consider two stars (or neutron stars, or black holes) each with mass  $M$ , separated by distance  $R$ , orbiting in the  $(x, y)$  plane. Using Newtonian gravity, the stars orbit with frequency

$$\omega^2 = \frac{2GM}{R^3} \quad (41)$$

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<sup>1</sup>We have assumed that the mass of the sun is approximately  $M_\odot \approx 2 \cdot 10^{30} \text{ kg}$

If we treat these stars as point particles, then the energy density is simply a product of delta-functions:

$$T^{00}(\mathbf{x}, t) = M\delta(z) \left[ \delta\left(x - \frac{R}{2} \cos(\omega t)\right) \delta\left(y - \frac{R}{2} \cos(\omega t)\right) + \delta\left(x + \frac{R}{2} \cos(\omega t)\right) \delta\left(y + \frac{R}{2} \cos(\omega t)\right) \right] \quad (42)$$

The quadrupole is defined as:

$$I_{ij}(t) = \int_{\Sigma} d^3x T^{00}(\mathbf{x}, t) x_i x_j \quad (43)$$

where  $\Sigma$  is the region where within there's the total energy. So it's easy to evaluate  $I_{ij}(t)$ :

$$I_{ij}(t) = \frac{MR^2}{2} \begin{pmatrix} \cos^2(\omega t) & \cos(\omega t) \sin(\omega t) & 0 \\ \cos(\omega t) \sin(\omega t) & \sin^2(\omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (44)$$

From here it is possible to derive the two polarisation states, that are:

$$h_+(t) = \frac{4}{r} \left( \frac{G\mathcal{M}_c}{c^2} \right)^{\frac{5}{3}} \left( \frac{\pi f_{GW}}{c} \right)^{\frac{2}{3}} \frac{1 + \cos^2(\theta)}{2} \cos(2\pi f_{GW} t_{ret} + 2\phi) \quad (45)$$

$$h_{\times}(t) = \frac{4}{r} \left( \frac{G\mathcal{M}_c}{c^2} \right)^{\frac{5}{3}} \left( \frac{\pi f_{GW}}{c} \right)^{\frac{2}{3}} \cos(\theta) \sin(2\pi f_{GW} t_{ret} + 2\phi) \quad (46)$$

where  $r$  is the distance,  $f_{GW}$  the frequency that we remember being time-dependent, the angle  $\theta = \pi/3$  and  $t_{ret} = t_c - t$  with  $t_c = 0$ . The phase  $\phi$  in this case was taken as zero. The gravitational waves can be polarised in different ways, and  $h_+$  and  $h_{\times}$  represent two distinct ways in which the wave can oscillate in space. Plotting only one of the two polarisations of gravitational waves, such as  $h_{\times}$ , or  $h_+$  as shown in Fig. (10), this means displaying the amplitude and phase of the gravitational waves in that specific polarisation. By plotting only one of the two polarisations, we are actually visualising the effect of the gravitational wave in a specific direction of observation. However, to get a complete picture of the behaviour of the gravitational waves in space, it would also be useful to consider a combination of the two polarisations, by plotting the sum or difference as shown in Fig. (11).

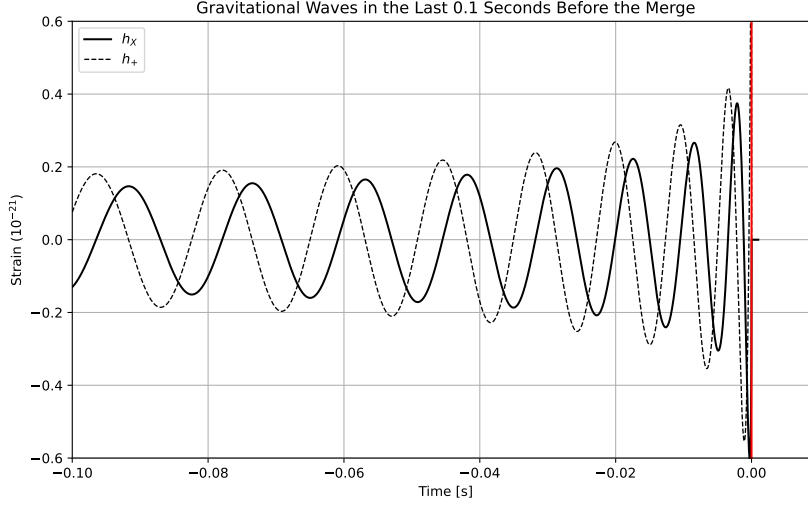


Figure 10: The curve of  $h_{\times}$  represents a gravitational wave oscillating in an inclined direction relative to the plane formed by the direction of propagation and the direction of observation in the polarisation  $h_{\times}$ . While the curve of  $h_{+}$  is in the polarisation  $h_{+}$ . The red line marks the coalescence time.

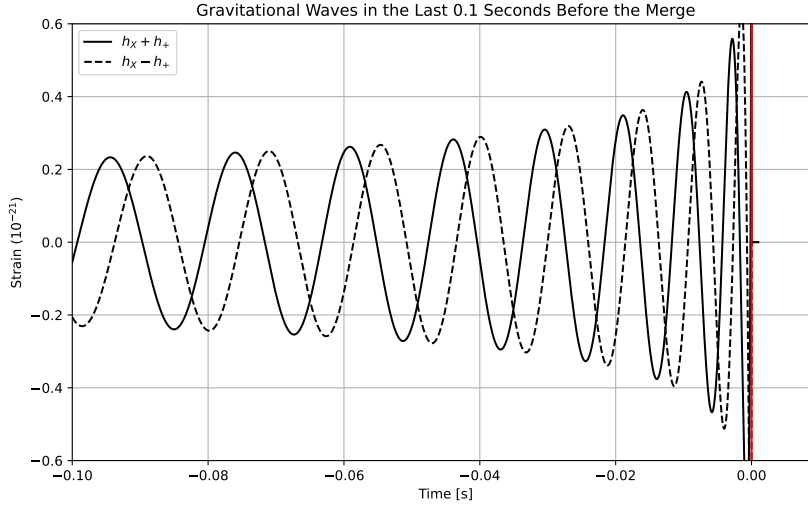


Figure 11: The sum of  $h_{+}$  and  $h_{\times}$  represents a linear combination of the two polarisations of gravitational waves. The difference between  $h_{+}$  and  $h_{\times}$  represents another linear combination of the polarisations of gravitational waves. The red line marks the coalescence time.

## Appendix A

```
import numpy as np
import matplotlib.pyplot as plt
import scipy.integrate as integrate

Omega_m0 = 0.3
h = 0.7
c = 299792.458 #km/s

def Hubble_law(z):
    return (c*z)/(100*h)

def integrand(z):
    E = (Omega_m0*(1+z)**3 + (1-Omega_m0))**(1/2)
    return 1/((1+z)*E)

def travel_time(zE):
    result, _ = integrate.quad(integrand, 0, zE)
    return c/(100*h)*result

def integrand1(z):
    E = (Omega_m0*(1+z)**3 + (1-Omega_m0))**(1/2)
    return 1/E

def comoving(zE):
    result, _ = integrate.quad(integrand1, 0, zE)
    return c/(100*h)*result

def luminosity(z):
    return (1+z)*comoving(z)

def angular(z):
    return comoving(z)/(1+z)

z = np.linspace(0, 100, 1000)

#Distances in Gpc
Hubble_distances = Hubble_law(z) / 1000
travel_times = np.array([travel_time(zi) for zi in z]) / 1000
comovings = np.array([comoving(zi) for zi in z]) / 1000
luminosities = np.array([luminosity(zi) for zi in z]) / 1000
angulars = np.array([angular(zi) for zi in z]) / 1000
```

```

plt.plot(z, Hubble_distances, label='Hubble Distance', color='black', linestyle='-')
plt.plot(z, travel_times, label='Travel Time Distance', color='black', linestyle='--')
plt.plot(z, comovings, label='Comoving Distance', color='black', linestyle='-.')
plt.plot(z, luminositys, label='Luminosity Distance', color='black', linestyle=':')
plt.plot(z, angulars, label='Angular Diameter Distance', color='black', linestyle='-')

plt.xlabel('Redshift')
plt.ylabel('Distance [Gpc]')
plt.title('Cosmological Distances vs Redshift')
plt.grid(True)
plt.xscale("log")
plt.yscale("log")
plt.legend()
plt.savefig('combined_plot.pdf')
plt.show()

```